

Talk Abstract

Constraining Dark Matter in the Era of Precision Astrometry: New Models for the Milky Way

The Hubble Space Telescope (HST) and Gaia have ushered a new era of high precision astrometry, providing accurate 3D velocity measurements of objects orbiting within the Milky Way's dark matter halo. When combined with detailed models of the MW's dark matter potential, such measurements become high-precision tools to constrain the dark matter mass profile of the MW. However, the MW hosts a massive satellite galaxy, the LMC, that changes the shape of the MW's dark matter potential in a non-symmetrical, time evolving manner. In this talk, I will demonstrate that high resolution simulations can be used to quantify the time evolving structure of the MW's dark matter potential owing to the influence of the LMC, enabling the orbit reconstruction of halo tracers that accurately accounts for the LMC's influence for the first time. The era of high-precision astrometry has arrived, yet we do not currently have an appropriate theoretical framework to study the assembly history of MW-like galaxies in the presence of massive satellite perturbers. The results of this work are critical to ongoing HST/JWST programs of the MW and broader efforts to understand the structure and evolution of the dark matter halo of other systems that host massive satellites.